Activity

Case Analysis – Avoiding Misleading Proportions

Question 1: Truncated Y-Axis in Sales Growth

A retail company presents the following monthly sales (in ₹ lakhs):

Month Sales

January 95

February 97

March 99

April 101

May 103

June 105

The marketing team creates a bar chart with the **Y-axis starting at 90 instead of 0**. The chart visually suggests that June sales are almost twice those of January.

Tasks:

1. Explain why the chart is misleading.

Answer:

The actual sales increased only from **₹95 lakhs to ₹105 lakhs**, which is about a **10.5% increase**.

But because the Y-axis starts at **90 instead of 0**, the bars appear much taller relative to each other, making June look almost twice as high as January. This exaggerates the actual growth.

2. Identify the visualization principle that has been violated.

Answer:

The chart violates the **principle of accurate and proportional representation**.

For bar charts, the Y-axis should generally start at **0** so that the lengths of the bars correctly represent the differences in values.

3. Recommend an improved visualization method.

Answer:

Improved visualization method

- Use a bar chart with the Y-axis starting at 0.
- Alternatively, use a line chart to clearly show the gradual monthly growth.
- Label the Y-axis as Sales (₹ lakhs) and show appropriate scale intervals.

4. Describe how this misleading chart could influence business decisions.

Answer:

Influence on business decisions

The misleading chart may make managers believe that sales have increased dramatically, when the actual increase is small. This could lead to:

- Overestimating business growth.
- Increasing production or inventory unnecessarily.
- Spending more on marketing based on exaggerated growth.
- Setting unrealistic future sales targets.
- Making incorrect investment or resource-allocation decisions.

Question 2: Pie Chart with Incorrect Proportions

A university reports student enrollment in four departments.

Department Students

Computer Science 420

Electronics 280

Mechanical 200

Civil 100

The designer accidentally creates a pie chart where the slices occupy approximately **50%, 30%, 15%, and 15%** instead of the correct proportions.

Tasks:

1. Calculate the correct percentage for each department.

Answer:

Correct percentage for each department

Total students = 420 + 280 + 200 + 100 = 1000

Department	Students	Correct Percentage
Computer Science	420	42%
Electronics	280	28%
Mechanical	200	20%
Civil	100	10%
Total	1000	100%

Calculation:

Percentage = (Department Students / Total Students) × 100

2. Explain why the displayed chart is misleading.

Answer:

The chart shows 50%, 30%, 15%, and 15%, which do not match the actual enrollment percentages of 42%, 28%, 20%, and 10%.

- Computer Science is exaggerated from 42% to 50%.
- Mechanical is reduced from 20% to 15%.
- Civil is exaggerated from 10% to 15%.
- The incorrect proportions can give viewers a false impression of department sizes.

Also, the displayed percentages add up to **110%**, which is impossible for a pie chart.

3. Suggest an alternative chart that would improve comparison

accuracy.

Answer:

Better alternative chart

A **bar chart** would be better because it allows users to compare department enrollment directly using the length of the bars.

A bar chart with:

- X-axis → Departments
- Y-axis → Number of Students

would make the differences between **420, 280, 200, and 100** much easier to see accurately.

4. Discuss how such errors could affect institutional planning.

Answer:

Effect on institutional planning

Incorrect visualization can lead to wrong decisions such as:

- Allocating too many classrooms or resources to a department.
- Hiring an inappropriate number of faculty members.
- Misallocating budgets.
- Incorrectly planning laboratories and infrastructure.
- Making inaccurate decisions about future admissions and department expansion.

Question 3: Unequal Bar Widths in Revenue Comparison

A software company compares annual revenue.

Product Revenue (₹ Crores)

A 60

B 75

C 90

D 105

Instead of using equal-width bars, the designer creates bars with increasing widths, making Product D appear nearly **three times larger** than Product A.

Tasks:

1. Identify the graphical error.

   Answer:

Identify the graphical error

The graphical error is **using unequal widths for the bars**. In a standard bar chart, all bars should have the **same width**. Only the height of the bars should represent the revenue.

2. Explain how unequal bar widths distort perception.

How unequal bar widths distort perception

Answer:

Unequal widths make viewers compare the **overall area** of the bars rather than just their heights.

The actual revenues are:

- Product A = ₹60 crores
- Product B = ₹75 crores
- Product C = ₹90 crores
- Product D = ₹105 crores

Product D's actual revenue is:

105 ÷ 60 = 1.75 times Product A's revenue.

So, D is not three times larger than A. Increasing the width of D makes its revenue appear much larger than it actually is, creating a **misleading visual impression**.

3. Redesign the visualization by specifying the correct design principles.

Answer:

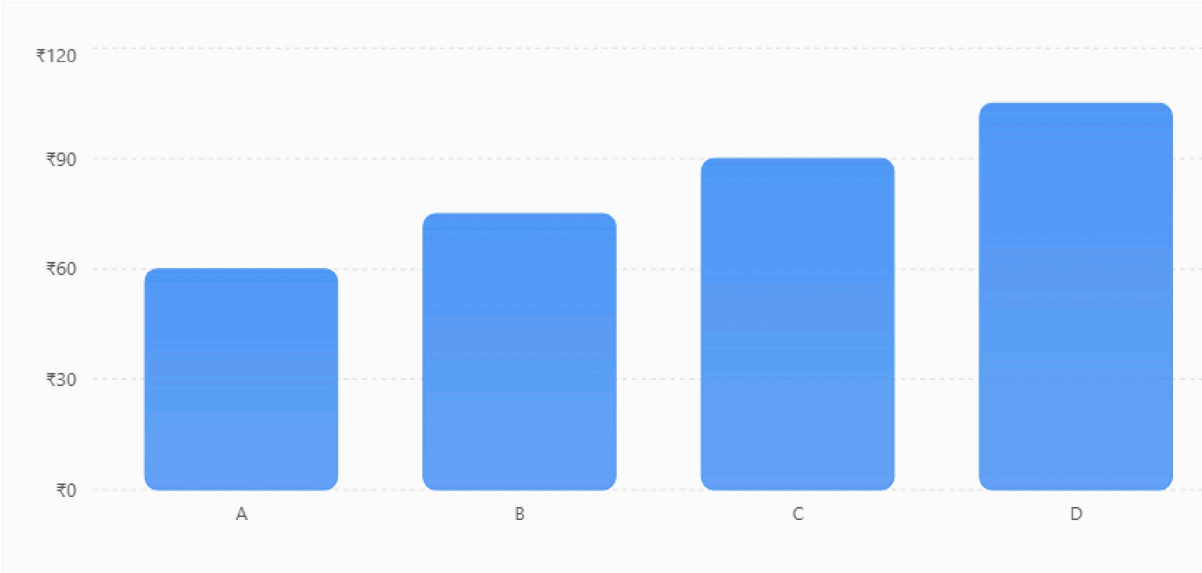
Redesign the visualization using correct principles

The corrected bar chart should follow these principles:

- Use **equal-width bars** for all products.
- Start the **Y-axis at 0**.
- Use a consistent Y-axis scale.
- Represent revenue only through **bar height**.
- Clearly label the X-axis as **Product**.
- Clearly label the Y-axis as **Revenue (₹ Crores)**.
- Avoid unnecessary 3D effects or other visual distortions.

Annual Revenue by Product

Revenue comparison using equal-width bars and a zero baseline.



4. Discuss the possible consequences if investors rely on this chart.

Answer:

Possible consequences if investors rely on this chart

A misleading chart can cause investors to:

- **Overestimate Product D's performance**.
- Invest more money based on an exaggerated visual impression.
- Misjudge the company's actual revenue growth.
- Make incorrect decisions about which products deserve investment.
- Lose confidence in the company when the actual financial data is examined.

Conclusion

The main problem is that bar width should remain constant. Revenue should be represented only by bar height, ensuring that the visualization gives an accurate and fair comparison.

any code required for that annual revenue

Question 4: 3D Column Chart Distortion

An energy company reports renewable energy generation.

Source Energy (GWh)

Solar 450

Source Energy (GWh)

Wind 500

Hydro 550

Biomass 520

The analyst uses a **3D column chart** with perspective effects. Due to the viewing angle, the Hydro bar appears almost **40% larger** than Wind, although the actual difference is only **10%**.

Tasks:

1. Explain why 3D charts can create misleading proportions.

   Answer:

3D charts use perspective, depth, and viewing angles. These effects can make bars in the front or at certain angles appear larger than they actually are.

In this case, the 3D perspective makes the Hydro bar look much larger than the Wind bar, even though their actual values are close.

2. Compare the perceived difference with the actual difference.

   Answer:

Wind generation = 500 GWh

Hydro generation = 550 GWh

Actual difference:

550 − 500 = 50 GWh

Percentage difference compared with Wind:

(50 ÷ 500) × 100 = 10%

So, Hydro is actually only 10% higher than Wind.

However, because of the 3D perspective, Hydro appears almost 40% larger. This is a significant visual distortion.

3. Recommend a better visualization.

Answer:

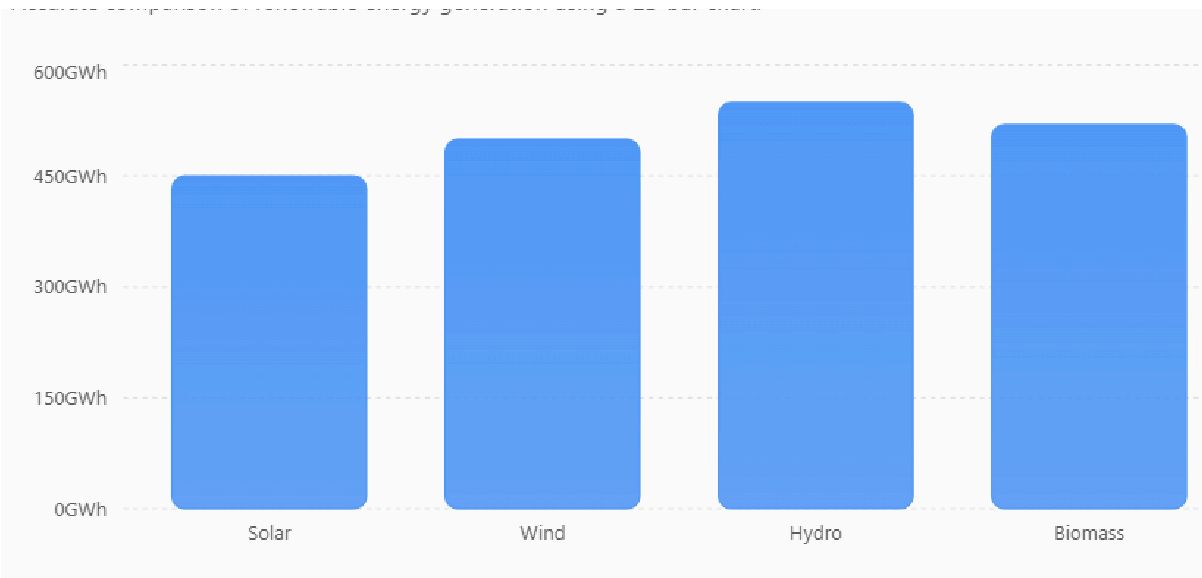
A **2D bar chart** is recommended because it provides a clear and accurate comparison.

It should have:

- Equal-width bars.
- A Y-axis starting at **0**.
- A consistent scale.
- No 3D perspective or unnecessary effects.
- Clear labels: Energy Source and Generation (GWh).

Renewable Energy Generation

Accurate comparison of renewable energy generation using a 2D bar chart.



4. Discuss the importance of avoiding visual distortion in sustainability reporting.

Answer:

Accurate visualization is important because sustainability reports are used by investors, management, regulators, and the public.

Misleading charts can:

- Give an incorrect impression of renewable energy performance.
- Lead to wrong investment and planning decisions.
- Make environmental progress appear better or worse than it really is.
- Reduce trust in the company's sustainability reports.
- Affect the company's credibility and transparency.

Conclusion: Sustainability data should be presented using simple, accurate 2D visualizations so that the information is not exaggerated or distorted.

Question 5: Bubble Chart with Incorrect Scaling

A healthcare organization compares disease cases across five

districts. **District Cases**

A 150

B 250

C 350

D 450

E 550

The analyst scales the **bubble radius directly** to the number of cases rather than scaling the **bubble area**. As a result, District E appears more than **ten times larger** than District A.

Tasks:

1. Explain why scaling the radius instead of the area is misleading.

   Answer:

In a bubble chart, people mainly perceive the area of the bubble, not just its radius.

If the radius is directly set equal to the number of cases, the bubble's area grows according to:

Area = πr^2

Therefore, increasing the radius causes the visible area to increase much faster than the actual data value.

For example:

- District A = 150 cases
- District E = 550 cases
- Actual ratio = 550 ÷ 150 ≈ 3.67

So District E has about 3.67 times as many cases as District A, not more than 10 times. Directly scaling radius exaggerates the difference.

2. Describe the correct method for proportional bubble sizing.

   Answer:

The bubble area should be proportional to the number of cases.

A suitable formula is:

Radius ∝ √(Number of cases)

This ensures that:

Bubble Area ∝ Number of cases

Thus, a district with twice as many cases should have approximately twice the bubble area, not twice the radius.

3. Suggest an alternative visualization suitable for this dataset.

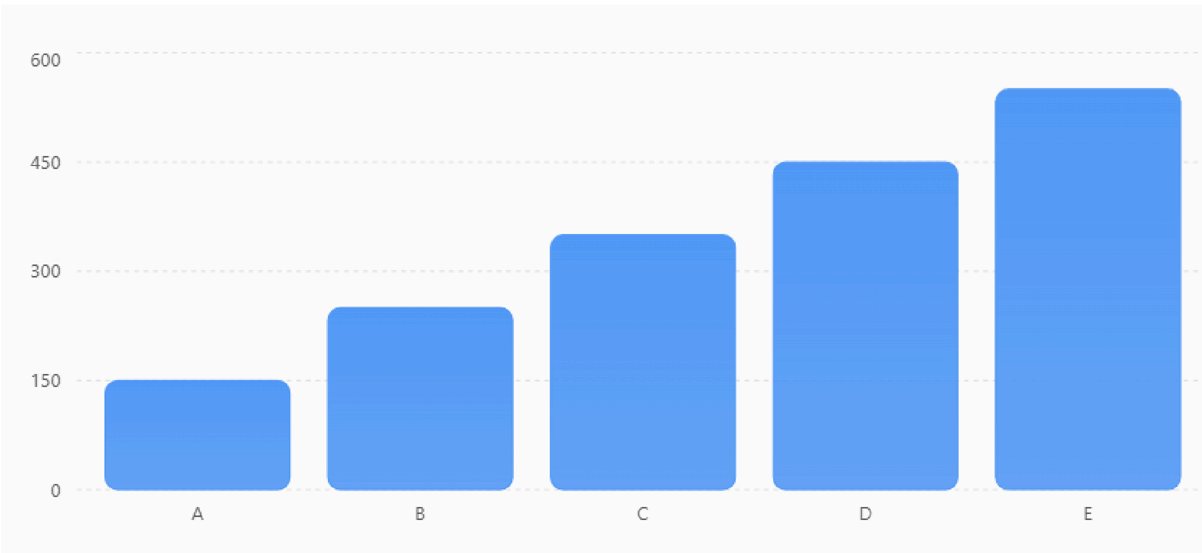
Answer:

Alternative visualization

A **bar chart** would be more suitable for this dataset because there are only five districts and the exact number of cases needs to be compared.

Disease Cases by District

Comparison of disease cases across five districts using a standard 2D bar chart.



A bar chart makes it easier to see the exact difference between districts.

4. Discuss the potential impact of this visualization on public health resource allocation.

Answer:

Impact on public health resource allocation

A misleading chart could cause authorities to:

- Allocate too many resources to districts that appear to have more cases.
- Allocate too few resources to districts with fewer cases.
- Misjudge the need for hospitals, medicines, and healthcare workers.
- Make incorrect decisions about vaccination or disease-control programs.
- Waste limited public health resources.

Conclusion: For accurate health-data comparison, a 2D bar chart is preferable. If a bubble chart is used, its area—not radius—must be proportional to the data value.